

Muhammad Ibtsaam Qadir

Address: 1395 Neil Armstrong Dr. Apt 20, West Lafayette, IN, USA

Email: ibtsaamqadir@gmail.com | Phone: +1 (326) 467 2769 | Web: ibtsaamqadir.github.io

Education

Ph.D. Biomedical Engineering

May 2024 – Present

Purdue University, West Lafayette, IN, USA

GPA: 3.87/4.00

Concentration: *Computational Biomedicine and Imaging*

Coursework: Data Mining, Quantitative Systems Biology, Medical Physics, Statistics for Health Sciences.

Graduate Certificate in Regulatory Affairs and Regulatory Science for Medical Devices

Aug 2026

Purdue University, West Lafayette, IN, USA

Bachelor of Electrical Engineering

Sep 2019 – Jun 2023

National University of Science and Technology (NUST), Islamabad, PK

GPA: 3.72/4.00

Coursework: Machine Learning, Signal Processing, Probability, Numerical Methods, Computer Vision.

Research Experience

Graduate Research Assistant

May 2024 – Present

Translational Medical Image Computing Lab, Weldon School of Biomedical Engineering, Purdue University, USA

- Developed **multimodal AI models** for pre-operative prediction of disease trajectories in pancreatic diseases using **3D medical imaging and clinical data**.
- **Curated and preprocessed large-scale multimodal datasets** including 3D medical imaging and clinical data for treatment planning and dynamic risk management in pancreatic surgery.
- Investigated the **effect of imaging acquisition parameters** in magnetic resonance imaging on the **performance of deep learning models**.
- Collaborating with international regulatory experts to analyze and **compare international regulatory guidelines for early-phase clinical investigations**.

Research Assistant

July 2023 – April 2024

NUST-Coventry Internet of Things Lab (NCIL), School of Electrical Engineering and Computer Science, NUST, PK

- Curated **large-scale datasets** and developed **AI predictive models to enhance environmental monitoring** accuracy.
- Drafted research proposals for international and national funding calls (Horizon Europe, UKRI).

Undergraduate Research Assistant

Feb 2020 – Feb 2021

COGNET Research Lab, School of Electrical Engineering and Computer Science, NUST, PK

- Designed a smart freight system integrating microcontrollers, Web 3.0, and automation for secure logistics operations.

Publications

Journal Publications

1. **Qadir, M. I.**, Baril J.A., Yip-Schneider M. T., et al. (2025). "Artificial Intelligence in pancreatic intraductal papillary mucinous neoplasm imaging: A systematic review." *PLOS Digital Health*
2. **Qadir, M. I.**, Hira, R. S., & Kolbinger, F. R. (2025). "From Marginal Gains to Clinical Utility: Machine Learning-Based Percutaneous Coronary Intervention Risk Prediction Models." *European Heart Journal-Digital Health*.
3. **Qadir, M. I.**, Mumtaz, R., Manzoor, M., et al. (2024). "Unmanned surface vehicle for intelligent water quality assessment to promote sustainable human health." *Water Supply*.

Accepted

4. **Qadir, M. I.**, Wober J., McDonnell A., et al. (2026). "EU's fragmented early-stage medical device study requirements need reform." *Nature Biomedical Engineering*.
5. Zhang, Z., **Qadir, M. I.**, Carstens, M., et al. (2026). "Prompt injection attacks on vision-language models for surgical decision support." *npj Digital Surgery*.

Conference Publications

6. Moshin, M., Umer, M., Bilal, A., **Qadir, M. I.**, et al. (2026). Channel Prediction under Network Distribution Shift Using Continual Learning-based Loss Regularization. *ICASSP*.

Preprints

7. **Qadir, M. I.**, Schonlau, D., Dydak U., Kolbinger, F. R. (2025). Field strength-dependent performance variability in deep learning-based analysis of magnetic resonance imaging. *arXiv*
8. Moshin, M., Umer, M., Bilal, A., Memon, Z., **Qadir, M. I.**, et al. (2025). On the Fundamental Limits of LLMs at Scale. *arXiv (Under review at Transactions on Machine Learning Research)*

Conference Abstracts

9. **Qadir M. I.**, Baril J.A., Yip-Schneider M. T., et al. (2025). “Towards personalized management of intraductal papillary mucinous neoplasms with multimodal artificial intelligence.” *American Society of Clinical Oncology (ASCO) Annual Meeting 2025*

Conference Presentations

1. Americas Hepato-Pancreato-Biliary Association (AHPBA) Annual Meeting 2026 (Talk). Prediction of Textbook Outcome after Distal Pancreatectomy Using Multimodal Artificial Intelligence Model.
2. Radiological Society of North America (RSNA) 111th Annual Meeting 2025 (Poster). “Beyond Tesla: Evaluating Deep Learning Generalization Across 1.5 T and 3.0 T MRI Scanners.”
3. IUSCCC Cancer Research Day 2025 (Poster). “Towards personalized management of intraductal papillary mucinous neoplasms with multimodal artificial intelligence.” (2nd Place for Best Poster Presentation)

Teaching Experience

Undergraduate Teaching Assistant

Sept 2022 – Jan 2023

School of Electrical Engineering and Computer Science (SEECS), NUST, Islamabad, PK
Course: CS114 – Fundamentals of Programming

Industry Experience

Technology Development Intern

Oct 2022 – Mar 2023

Synaptic Delver, SG

- Developed methods for exploring and analyzing biosignals for improved signal interpretation.
- Implemented Web 3.0 solutions for secure storage of neurological signals, enhancing data privacy and security.

Printed Circuit Board Design Intern

July 2021 – Sep 2021

RADAR Lab, School of Interdisciplinary Engineering & Science (SINES), NUST, PK

- Designed digital printed circuit boards using Altium Designer and Saturn PCB Toolkit, including a DC-DC 5V converter with Intel Enpirion IC, optimized for thermal efficiency.

Grants

IUSCCC Trainee Pilot Grant, for the project titled “Radiologist-AI Collaboration for Early Detection of Pancreatic Cancer: Optimizing Interaction in IPMN Management”

Awards and Honors

1. Leslie Bottorff Fellowship Awardee, 2025
2. Purdue Graduate Student Government Travel Award, 2025
3. 2nd Place for Best Poster Presentation, IUSCCC Cancer Research Day, 2025
4. Prototype Development Grant, Finding Innovation & Creative Solutions (FICS) Competition, 2024.
5. Merit Scholarship, Fazaia Educational Welfare Scheme, PK, 2019 – 2023.
6. Merit Award, Federal Board of Intermediate and Secondary Education, PK, 2017 – 2019, 2019 – 2023.

Mentorship

Graduate Student Research Mentor

Aug 2024 – Present

Weldon School of Biomedical Engineering, Purdue University

Mentees: *Steven Kiss (Undergrad)*, *Maria Moreano (Undergrad)*, *Madison Loiselle (Undergrad)*, *Saloni Jajoo (Undergrad)*, *Maksymilian Mroczkowski (Summer Undergraduate Research Fellow-Imperial College of London)*

Professional Memberships

Trainee Member

Sept 2024 – Present

Indiana University Melvin and Bren Simon Comprehensive Cancer Center (IUSCCC), Indianapolis, IN, USA

Peer Review

Journals

Radiology: Artificial Intelligence, Radiology: Cancer Imaging, International Journal of Surgery, npj Precision Oncology, Scientific Data, PLOS Digital Health, Journal of Magnetic Resonance Imaging, European Radiology

Conferences

Medical Image Understanding and Analysis (MIUA) Conference (2024, 2025)